



Article

Experimental Investigation on Mechanical Properties of Glass Fiber–Nanoclay–Epoxy Composites Under Water-Soaking: A Comparative Study Using RSM and ANN

Manjunath Shettar ¹, Ashwini Bhat ^{2,*}, Nagaraj N. Katagi ² and Mandya Channegowda Gowrishankar ¹

¹ Department of Mechanical and Industrial Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal 576104, Karnataka, India; gowri.shankarmc@manipal.edu (M.C.G.)

² Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal 576104, Karnataka, India

* Correspondence: ashwini.bhat@manipal.edu

Abstract: Fiber-reinforced polymer composites are exposed to severe environmental conditions throughout their intended lifespan. It is essential to investigate how they age when exposed to cold and hot water to increase the durability of fiber-reinforced polymer composites. This work uses a hand lay-up process to create composites with different weight percentages of glass fiber, nanoclay, and epoxy. ASTM guidelines are followed for performing tensile and flexural tests. The input parameters, varying wt.% of glass fiber and nanoclay, are continuous, and the aging condition is deemed a categorical factor. The mechanical properties are considered as response variables (output). The mechanical properties are optimized using Response Surface Methodology (RSM), while Artificial Neural Networks (ANNs) provide a reliable predictive model with high correlation coefficients. The findings demonstrate that ANNs outperform RSM in flexural strength prediction, whereas RSM offers greater accuracy for tensile strength modeling. SEM analysis of the fracture surfaces reveals the causes of specimen failure under tensile load, with distinct differences between dry, cold, and boiling water-soaked specimens.

Keywords: composite materials; mechanical properties; materials testing; composite manufacturing; RSM; ANN; SEM



Academic Editor: Giuseppe Cavallaro

Received: 20 March 2025

Revised: 16 April 2025

Accepted: 19 April 2025

Published: 21 April 2025

Citation: Shettar, M.; Bhat, A.; Katagi, N.N.; Gowrishankar, M.C. Experimental Investigation on Mechanical Properties of Glass Fiber–Nanoclay–Epoxy Composites Under Water-Soaking: A Comparative Study Using RSM and ANN. *J. Compos. Sci.* **2025**, *9*, 195. <https://doi.org/10.3390/jcs9040195>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Fiber-reinforced polymer (FRP) composites are now widely considered for structural and engineering applications based on their high mechanical properties, low weight, and corrosion resistance [1,2]. Among them, glass fiber-reinforced polymer (GFRP) composites have found large-scale applications in the aerospace, automobile, naval, and building construction industries due to their superior strength-to-weight ratio, high impact resistance, and ease of manufacture [3,4]. However, they are frequently questioned in terms of the environmental influences on their durability over the long term, primarily by moisture exposure and water submersion. Under water-soaking conditions, GFRP composites undergo a multi-stage moisture uptake process involving the diffusion of water molecules through the polymer matrix and along the fiber–matrix interface. The absorbed moisture deteriorates the mechanical properties of GFRP through fiber–matrix debonding, matrix plasticization, and hydrothermal aging [5–7]. It is thus crucial for the reliable utilization of GFRP composites in a wet environment to understand how water absorption alters the mechanical characteristics of GFRP composites.

Different nanofillers have been studied to improve the performance and long-term durability of epoxy-based GFRP composites, with nanoclay being a key reinforcement material [8]. Dispersal of nanoclay particles in the polymer matrix significantly enhances barrier characteristics, thermal stability, and mechanical performance [9,10]. Nanoclay addition increases the resistance of composites against moisture penetration by lowering the permeability of the polymer matrix. In addition, nanoclay has the potential to enhance interfacial adhesion between the fibers and the matrix, thus reducing the detrimental impact of water exposure on mechanical integrity [8,11]. Notwithstanding these benefits, the effect of nanoclay on the water absorption properties and the resultant mechanical behavior of GFRP composites subjected to extended soaking conditions is an area of active research.

Optimization of the mechanical properties of fiber-reinforced composites (FRCs) is important for maximizing their performance in structural applications. Response Surface Methodology (RSM) and Artificial Neural Networks (ANNs) have proven to be effective tools for modeling and predicting the behavior of FRCs under different conditions [12]. RSM, a statistical tool, allows for the creation of empirical relationships between input parameters and mechanical properties, enabling optimization with a reduced number of experimental trials [13]. Conversely, an ANN, an artificial learning method, performs optimally in capturing intricate nonlinear patterns in huge databases with high-accuracy predictions [14]. Integration of RSM and ANNs makes the mechanical property estimations more reliable, contributing to better material design and performance improvement.

An article by Arunachalam et al. [15] examined the impact of silane treatment on the performance of hybrid composite through RSM and ANNs. Their paper determines the best proportion of nanoparticles, silane amount, and dipping time for improving flexural strength and hardness. Their results show that silane treatment greatly enhances fiber–matrix bonding, with the ANN model having 95% accuracy. Their study is of great significance to improving composite properties for structural use, showing the efficiency of both RSM and ANNs in simulating mechanical behavior. In another work, Arunachalam et al. [16] discussed the mechanical properties of jute/kenaf/glass fiber composites reinforced with multi-walled carbon nanotubes (MWCNTs) using RSM and ANNs for improvement. RSM is used to investigate the simultaneous effect of fiber orientation, sequencing, and MWCNT content on hardness and flexural strength. The research successfully integrates RSM and ANNs to maximize performance with a strong method of composite material design.

The article by Ladaci et al. [17] applied ANN and RSM methods to model the behavior of uptake measured in experiments and optimize the immersion period and PWW fiber content in RHDPE/PWW biocomposite WU. Findings show that the ANN models' training, test, and validation correlation coefficients were 0.9984, 0.9955, and 0.9723, respectively, for predicting WU. The RSM's model correlation coefficients are around 96.24% and 95.63%, respectively. Concerning accuracy and reliability, the ANN model outperformed the RSM model, making it suitable for various industrial uses.

This study presents an experimental investigation into the mechanical properties of glass fiber–nanoclay–epoxy composites with varying wt.% of glass fiber and nanoclay. The composites are subjected to different water-soaking conditions, viz., cold and boiling water-soaking. The input parameters, varying wt.% of glass fiber and nanoclay, are continuous factors, and the aging conditions are deemed a categorical factor. The mechanical properties are considered as response variables (output). The research employs RSM and ANNs to model and predict the mechanical properties of water-aged composites. By comparing these two approaches, this study aims to provide insights into each method's predictive capability and reliability in evaluating the mechanical behavior of water-aged composites.

2. Materials and Methods

2.1. Materials and Preparation of Composites

Epoxy resin (L-12) and hardener (K-6) are procured from “Atul Polymers, Gujarat, India”, while the bi-directional woven E-glass fabric reel is sourced from “Yuje Enterprises, Bengaluru, India”. Additionally, nanoclay (surface-modified with 25–30 wt.% trimethyl stearyl ammonium) is obtained from “Sigma Aldrich”, Bengaluru, India. Table 1 presents some properties of the materials used for the present study.

Table 1. Properties of materials used for the present study.

Material Name	Properties
Glass fiber—Plain bi-directional woven, 360 GSM	Tensile strength—1720–1950 MPa Tensile modulus—72–76 GPa Density—2.5 g/cm ³
Epoxy resin (L-12) and hardener (K-6) (Mixing ratio 10:1)	Tensile strength—55–70 MPa Tensile modulus—2.5–04 GPa Flexural strength—120–140 MPa Density—1.15 g/cm ³ Appearance (color)—White to off-white Appearance (form)—Powder
Nanoclay (Montmorillonite (MMT))	Size—<20 µm Bulk density—200–500 kg/m ³ Surface-modified contains 15–35 wt.% octadecylamine, 0.5–5 wt.% aminopropyltriethoxysilane.

Nanoclay with varying wt.%, as mentioned in Table 2, is dispersed in epoxy using a magnetic stirrer for 30 min, followed by 20 min of sonication. The nanoclay–epoxy mix and hardener are blended thoroughly. Composite laminates with varying glass fiber wt.%, as mentioned in Table 2, are prepared using the hand lay-up technique, followed by compression molding. A roller is used to roll on wet laminates to compact the laminate layers and push air bubbles out. Rolling is completed in multiple directions, starting from the center and moving outward to ensure an even pressure distribution. Compression molding is carried out at 100 Bar and 50 °C for a period of 24 h. The dimensions of the laminates are 300 × 300 × 3 mm³. With a 1% bilateral tolerance, the volume is kept constant. They are compressed to keep the laminates equal to 3 mm thickness by a stopper. Every laminated composite has a thickness of 3 mm. Composite specimens are cut from cured laminate according to ASTM standards for testing.

Table 2. Factors and different levels used in the present study.

Factors	Level 1	Level 2	Level 3
Nanoclay (NC) wt.%	0	2	4
Glass fiber (GF) wt.%	40	50	60
Aging conditions (AG)	Dry Specimens (DS)	Cold water-soaking (CS)	Boiling water-soaking (BS)

2.2. Water-Soaking Conditions

The water-soaking conditions are aimed to simulate long-term and accelerated hydrothermal aging conditions that polymer composites may experience in real-world environments. The specimens are kept in the water for 60 days at ambient temperature for cold water-soaking. Cold water-soaking represents prolonged exposure to humid or submerged conditions, allowing for gradual water uptake and its effects on the composite structure

to be assessed over time. Meanwhile, for boiling water-soaking, the specimens are kept in boiling water for 2 h and left in the water for 22 h. These processes are repeated each day for 10 days, leading to a total boiling time of 20 h [18]. Boiling water-soaking is an accelerated aging method that mimics extreme environmental stress by enhancing moisture diffusion and potential degradation mechanisms.

2.3. Mechanical Testing and Analysis

2.3.1. Tensile Test

The ZWICK ROELL automated Universal Testing Machine (UTM) is used to conduct tensile tests on composite materials following the ASTM D3039 standard [19]. This ensures comprehensive evaluation while maintaining standardized procedures for reliable results. The testing is performed at a jaw speed of 1 mm/min. Five specimens of each set and condition are tested, and the average value is considered tabulation and discussion.

2.3.2. SEM Analysis

SEM analysis is conducted using a ZEISS Scanning Electron Microscope (SEM) following the tensile test. The specimen is cut to the required size and securely mounted for examination. Before analysis, a thin conductive coating is applied using a sputter coater, which takes approximately 10 min to complete.

2.3.3. Flexural Test

The ZWICK ROELL automated Universal Testing Machine (UTM) is used to conduct three-point flexural tests on composite materials following the ASTM D7264 standard [20]. A span-to-thickness ratio of 32:1 is considered for the testing. The jaw speed for this test is set at 2 mm/min. Five specimens of each set and condition are tested, and the average value is considered tabulation and discussion.

2.4. Design of Experiments

The Central Composite Design of RSM is employed to minimize the number of required trials, reduce cost, and save time while ensuring robust statistical analysis. The methodology enables the evaluation of both individual and interaction effects of various factors (at different levels) on mechanical properties, viz., tensile and flexural strengths. The input parameters, varying wt.% of glass fiber and nanoclay, are continuous factors, and the aging conditions are deemed a categorical factor.

Minitab® 22 software is utilized to create the matrix design and analyze the experimental data. Three levels for each factor, viz., glass fiber wt.%, nanoclay wt.%, and water-soaking conditions, are considered, as shown in Table 2.

2.5. Artificial Neural Network

An ANN is utilized to predict the mechanical properties (tensile strength and flexural strength) of the glass fiber–nanoclay–epoxy composites subjected to different aging conditions. The ANN model is developed to capture the nonlinear relationships between the input variables (nanoclay wt.%, glass fiber wt.%, and aging conditions) and the output responses. The architecture used in this study is shown in Figure 1.

The ANN model consists of three primary layers. The input layer contains three neurons corresponding to the input parameters nanoclay wt.%, glass fiber wt.%, and aging conditions. The hidden layer comprises a single layer with ten neurons, optimized based on preliminary trials to balance computational efficiency and accuracy. The output layer includes two neurons representing the predicted mechanical properties: tensile strength and flexural strength. The input and output neurons are normalized to improve the

convergence of the training process. A hyperbolic tangent sigmoid (tansig) activation function is used in the hidden layer, while a linear function is applied to the output layer.

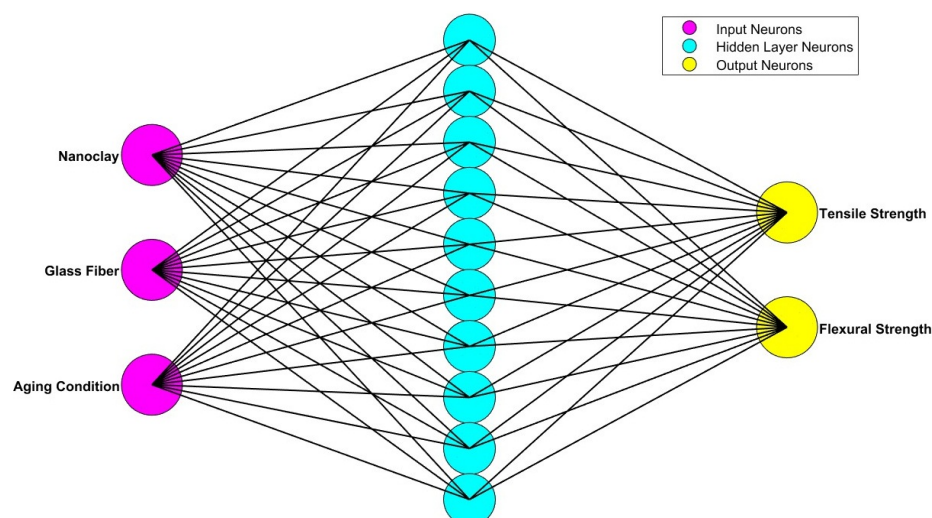


Figure 1. ANN architecture.

The ANN model is developed and implemented using MATLAB 2024a coding, and the model is trained using the backpropagation algorithm with the Levenberg–Marquardt optimization technique. The dataset is split into three subsets: 70% for training, 15% for validation, and 15% for testing. The training set is used to develop the model, while the validation set helps fine-tune the parameters and prevent overfitting. The testing set assesses the generalization capability of the trained model.

3. Results and Discussion

3.1. Response Surface Methodology—Central Composite Design

Table 3 displays the Central Composite Design, and 42 tests are conducted in total. The response variables for this current analysis are tensile and flexural strengths

Table 3. Central Composite Design with levels, factors, and response variables.

Nanoclay (NC) wt.%	Glass Fiber (GF) wt.%	Aging Conditions (AG)	Tensile Strength (MPa)	Flexural Strength (MPa)
4	60	CS	295	388
2	50	DS	283	402
4	60	BS	290	373
2	50	BS	227	344
0	40	BS	190	275
4	40	CS	234	349
2	50	CS	235	355
2	50	CS	247	366
2	50	DS	280	414
2	50	BS	233	354
0	40	CS	199	289
4	40	DS	281	399
0	60	BS	235	315
2	50	DS	285	378
0	60	DS	286	375
0	60	CS	243	327

Table 3. Cont.

Nanoclay (NC) wt.%	Glass Fiber (GF) wt.%	Aging Conditions (AG)	Tensile Strength (MPa)	Flexural Strength (MPa)
4	60	DS	319	429
2	50	CS	241	334
0	40	DS	255	341
4	40	BS	233	338
2	50	BS	227	324
2	50	DS	275	390
4	50	CS	255	373
2	50	CS	223	344
4	50	BS	244	360
2	60	DS	290	418
2	50	BS	216	334
0	50	BS	205	298
2	40	BS	199	325
0	50	CS	208	311
2	50	CS	230	377
4	50	DS	291	416
2	50	CS	220	360
2	50	BS	221	365
2	50	BS	210	340
0	50	DS	252	360
2	60	CS	254	369
2	60	BS	243	359
2	50	DS	276	426
2	50	DS	269	417
2	40	CS	212	337
2	40	DS	263	391

3.1.1. ANOVA

Analysis of Variance (ANOVA) is a statistical method used to test hypotheses. In ANOVA, the primary emphasis is on the “*p*-Value” column, which helps determine statistical significance. A *p*-value threshold of 0.05 (corresponding to a 95% confidence level) is commonly used. If the *p*-value is below 0.05, it indicates a statistically significant effect, whereas a value above 0.05 suggests that the effect is not significant. As shown in Table 4, all linear terms (nanoclay, glass fiber, and aging conditions) and interaction terms (nanoclay*glass fiber, nanoclay*aging conditions, and glass fiber*aging conditions) significantly influence tensile strength. Table 5 presents all linear terms (nanoclay, glass fiber, and aging conditions) and square terms (nanoclay²) that significantly affect flexural strength. Typically, an R-squared (R²) value between 90% and 100% indicates a highly correlated model. In these analyses, the R²(pred) values are 96.89% and 91.17%, demonstrating a strong correlation with the experimental data and confirming the model’s overall significance.

Table 4. ANOVA table for tensile strength.

Source	DF	Adj SS	Adj MS	F-Value	p-Value
Model	12	40,291.4	3357.6	163.21	0.000
Blocks	1	1108.8	1108.8	53.90	0.000
Linear	4	37,919.7	9479.9	460.80	0.000
Nanoclay	1	7564.5	7564.5	367.70	0.000
Glass fiber	1	8406.7	8406.7	408.64	0.000
Aging conditions	2	21,948.4	10,974.2	533.44	0.000
Square	2	188.9	94.5	4.59	0.019
Nanoclay*Nanoclay	1	39.2	39.2	1.91	0.178
Glass fiber*Glass fiber	1	83.2	83.2	4.04	0.054
2-Way Interaction	5	548.4	109.7	5.33	0.001
Nanoclay*Glass fiber	1	108.0	108.0	5.25	0.029
Nanoclay* Aging conditions	2	157.0	78.5	3.82	0.034
Glass fiber* Aging conditions	2	283.4	141.7	6.89	0.004
Error	29	596.6	20.6		
Lack-of-Fit	17	345.9	20.3	0.97	0.532
Pure Error	12	250.7	20.9		
Total	41	40,888.0			
S=4.53571	R-sq=98.56%	R-sq(adj)=97.94%	R-sq(pred)=96.89%		

Table 5. ANOVA table for flexural strength.

Source	DF	Adj SS	Adj MS	F-Value	p-Value
Model	12	53,294.7	4441.2	34.87	0.000
Blocks	1	231.3	231.3	1.82	0.188
Linear	4	50,071.7	12,517.9	98.28	0.000
Nanoclay	1	15,842.0	15,842.0	124.38	0.000
Glass fiber	1	5304.5	5304.5	41.65	0.000
Aging conditions	2	28,925.2	14,462.6	113.55	0.000
Square	2	2194.8	1097.4	8.62	0.001
Nanoclay*Nanoclay	1	1781.4	1781.4	13.99	0.001
Glass fiber*Glass fiber	1	12.9	12.9	0.10	0.753
2-Way Interaction	5	66.3	13.3	0.10	0.990
Nanoclay*Glass fiber	1	5.3	5.3	0.04	0.839
Nanoclay* Aging conditions	2	25.0	12.5	0.10	0.907
Glass fiber* Aging conditions	2	36.0	18.0	0.14	0.869
Error	29	3693.7	127.4		
Lack-of-Fit	17	239.0	14.1	0.05	1.000
Pure Error	12	3454.7	287.9		
Total	41	56,988.4			
S=11.2858	R-sq=93.52%	R-sq(adj)=90.84%	R-sq(pred)=91.17%		

3.1.2. Regression Equation and Residual Plots

Regression equations for tensile and flexural strengths are obtained according to categorical factors, i.e., water-soaking conditions. Equations (1)–(3) are used to predict composites' tensile strength, and Equations (4)–(6) are used to predict composites' flexural strength, respectively, for water-soaking conditions.

$$\text{Tensile strength of dry specimens} = 277.2 - 1.52(NC) - 1.89(GF) + 0.547(NC^2) + 0.0319(GF^2) + 0.1500(NC \times GF) \quad (1)$$

$$\text{Tensile strength of cold water – soaking specimens} = 185.2 + 1.48(NC) - 1.04(GF) + 0.547(NC^2) + 0.0319(GF^2) + 0.1500(NC \times GF) \quad (2)$$

$$\text{Tensile strength of boiling water – soaking specimens} = 176.7 + 1.73(NC) - 1.06(GF) + 0.547(NC^2) + 0.0319(GF^2) + 0.1500(NC \times GF) \quad (3)$$

$$\text{Flexural strength of dry specimens} = 250 + 30.42(NC) + 2.84(GF) - 3.689(NC^2) - 0.0126(GF^2) - 0.033(NC \times GF) \quad (4)$$

$$\text{Flexural strength of cold water – soaking specimens} = 185 + 31.67(NC) + 3.14(GF) - 3.689(NC^2) - 0.0126(GF^2) - 0.033(NC \times GF) \quad (5)$$

$$\text{Flexural strength of boiling water – soaking specimens} = 172 + 31.67(NC) + 3.14(GF) - 3.689(NC^2) - 0.0126(GF^2) - 0.033(NC \times GF) \quad (6)$$

Figures 2 and 3 illustrate that the residuals are closely aligned with the fitted axis, with only slight deviations from a normal distribution. This indicates that the residuals follow a normal dispersion pattern, reinforcing a strong linear relationship between the factors and the response variable. In all other graphs, the residuals are randomly scattered, which is essential for ensuring a good fit between the experimental and predicted values.

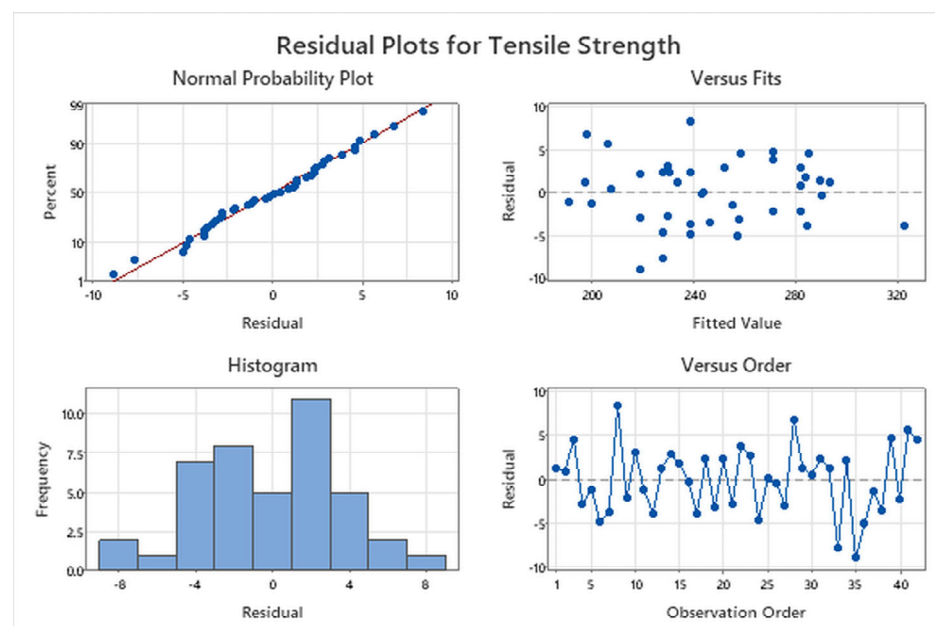


Figure 2. Residual plots for tensile strength.

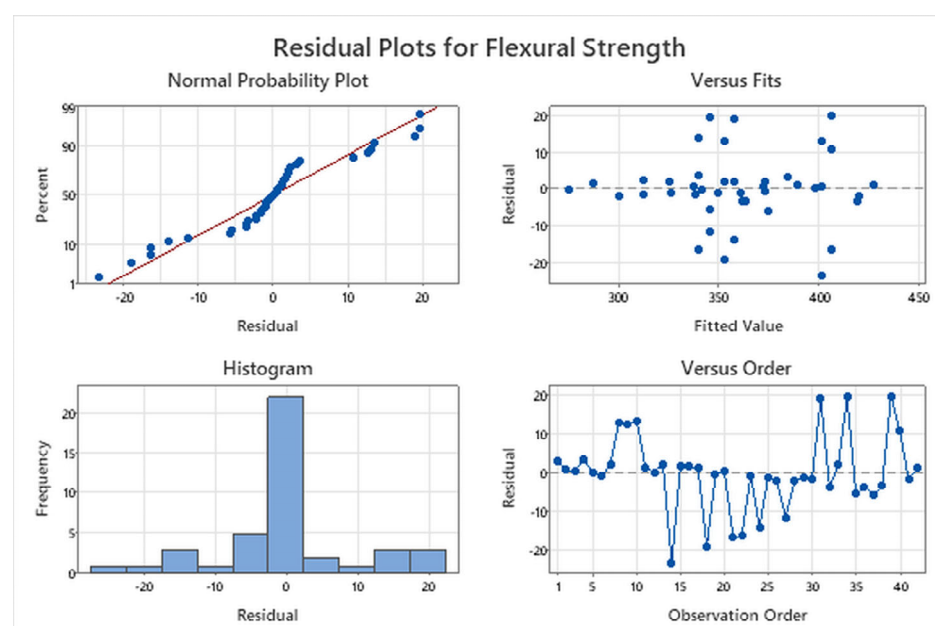


Figure 3. Residual plots for flexural strength.

3.1.3. Main Effects Plots

Figure 4 offers important information regarding the effect of nanoclay wt.%, glass fiber wt.%, and aging conditions on the tensile strength of the composite material. The main effect plots indicate how these factors singly contribute to the mechanical performance of the composite, providing a clear picture of their influence.

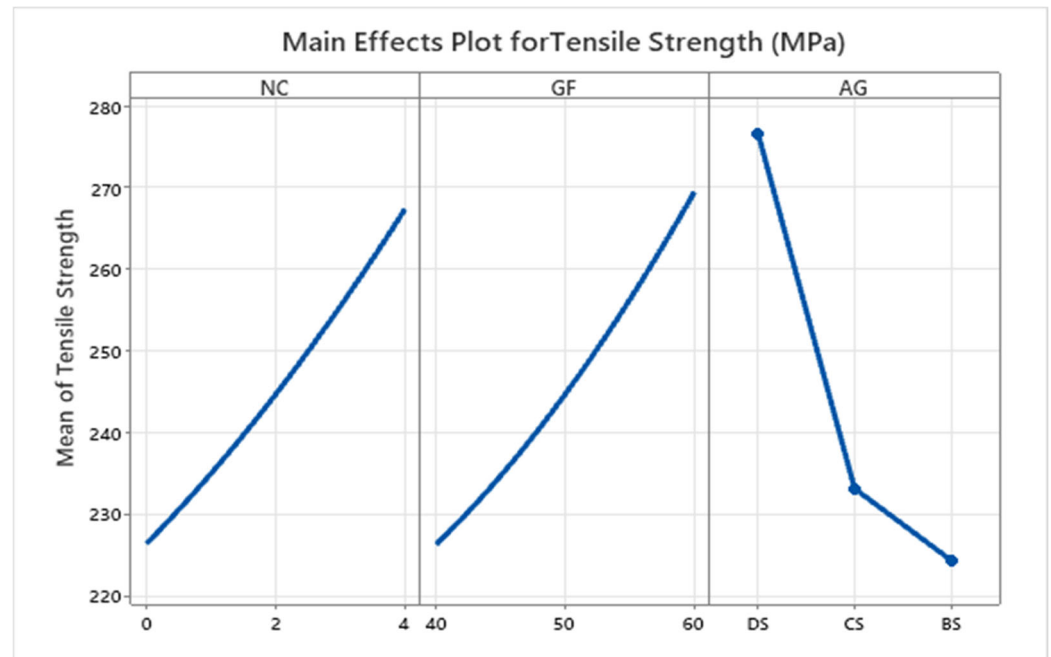


Figure 4. Main effect plot for tensile strength.

Also, the effect of nanoclay wt.% is seen because the tensile strength improves steadily with the rise in nanoclay from 0 wt.% to 4 wt.%. The good reinforcing qualities of nanoclay can explain this significant improvement in enhancing interfacial bonding between reinforcements and the polymer matrix [21,22]. Nanoclay, with its high surface area and layered nature, enhances the transfer of stress within the composite and renders it more resistant to mechanical deformation [23]. The increasing trend in tensile strength again confirms that nanoclay is responsible for the improvement in the structural strength of fiber-reinforced polymer composites.

Similarly, the glass fiber wt.% exhibits a substantial positive effect on tensile strength, as observed in the increasing trend from 40 wt.% to 60 wt.%. Glass fiber is a well-established reinforcement material that significantly improves the load-bearing capacity of composites. The composite structure becomes more robust as its wt.% increases, leading to better mechanical performance [24]. This behavior is consistent with basic composite mechanics, wherein greater reinforcement typically increases strength and stiffness [25]. The increasing tensile strength with increasing glass fiber content supports its efficacy in strengthening polymer-based materials.

On the other hand, the effect of aging conditions shows a sharp contrast. The tensile strength is highest for dry specimens (as-made specimens). There is a steep fall when the composite is immersed in cold water-soaking conditions, and a more significant drop is noted when immersed in boiling water-soaking conditions. This degradation is due to moisture absorption, which weakens the fiber–matrix interface and facilitates microcracks formation [26]. The hydrothermal aging impacts in boiling water further enhance polymer degradation, resulting in extensive deterioration of tensile strength [27].

Figure 5 displays the main effects plots for flexural strength and gives essential information regarding the effects of nanoclay wt.%, glass fiber wt.%, and aging conditions

on the flexural strength of the composite material. This figure shows how each factor individually contributes to the flexural strength, clearly comprehending their individual effects on composite performance.

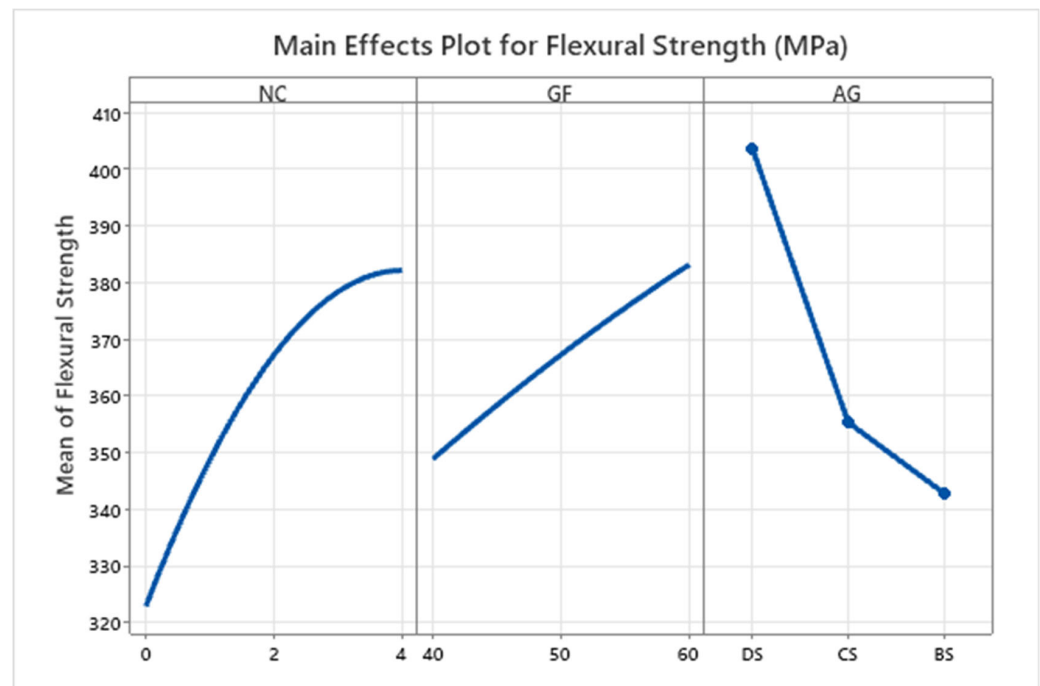


Figure 5. Main effect plot for flexural strength.

Beginning with wt.% nanoclay, the flexural strength follows a steeply rising trend between 0 wt.% and 2 wt.%, followed by a slight leveling off as content increases to 4 wt.%. This indicates that nanoclay strongly enhances the bending stress resistance of the composite by reinforcing the polymer matrix and facilitating load transfer between fibers [28]. Yet, the marginal gain after 2 wt.% suggests that too much nanoclay can cause agglomeration, which might decrease the efficiency of stress transmission within the material. However, the overall enhancement in flexural strength confirms the reinforcing ability of nanoclay in fiber–polymer composites.

A similar strengthening effect is observed with glass fiber wt.%, where the flexural strength increases as the glass fiber content rises from 40 wt.% to 60 wt.%. This trend is expected, as glass fibers are known for their excellent stiffness and strength, significantly improving the composite’s resistance to flexural deformation. Higher fiber content results in better load-bearing capacity, reinforcing the material’s ability to withstand bending forces [27]. The steady increase in flexural strength confirms the well-established role of glass fibers in enhancing the mechanical performance of polymer composites.

However, the aging conditions reveal a contrasting impact, with dry specimens exhibiting the highest flexural strength. A substantial decline is observed when specimens undergo cold water-soaking conditions, and a more pronounced reduction occurs in boiling water-soaking conditions. The sharp drop in flexural strength due to moisture exposure is attributed to water absorption, which degrades the fiber–matrix interface, causing delamination and reducing load transfer efficiency [5]. Boiling water further accelerates the degradation process, weakening the composite’s ability to resist flexural stress [29]. This decline highlights the significant effect of hydrothermal aging on composite durability and mechanical stability.

3.1.4. Interaction Plots

Figure 6 provides the interaction plot for the tensile strength and a close observation of how various factors interact and affect the tensile strength of a composite material. The interaction of NC and GF tells us that tensile strength becomes better as the content of glass fiber increases for every level of nanoclay. This is caused by the reinforcement of glass fibers, which increases the load-carrying capability of the composite. Additionally, nanoclay offers higher interfacial adhesion between the matrix and fiber. However, the improvement rate in the tensile strength seems more significant at higher levels of GF, indicating a synergistic interaction between nanoclay and glass fiber.

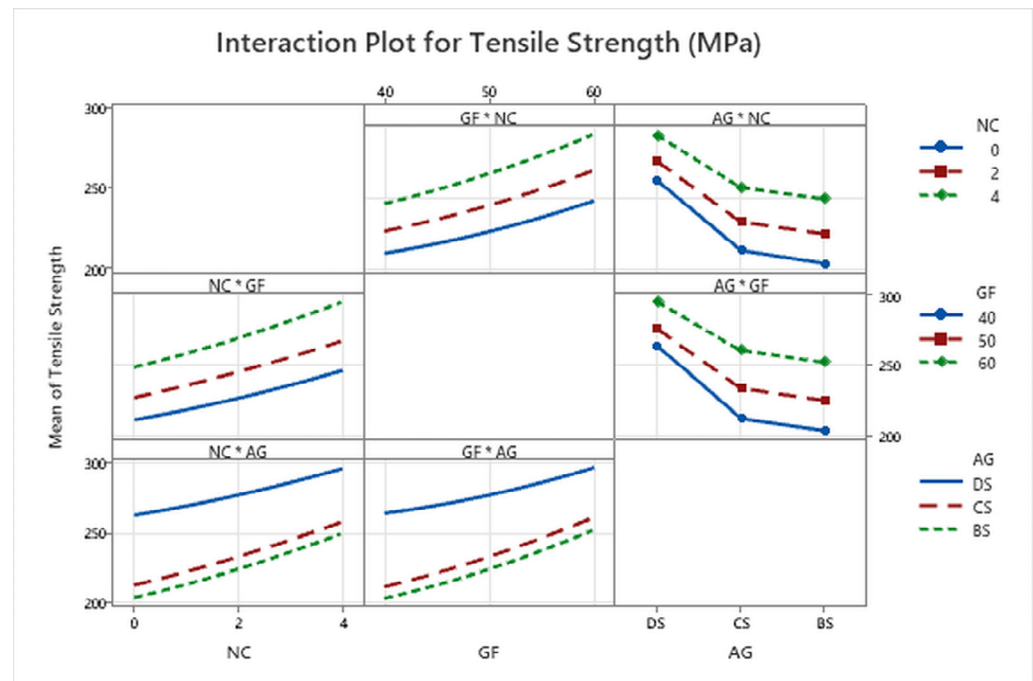


Figure 6. Interaction plot for tensile strength.

The interaction of AG and NC also shows the influence of aging conditions on the mechanical properties of the composite. The aging conditions negatively affect tensile strength. Cold and boiling water-soaking conditions lead to moisture-induced plasticization, which decreases matrix stiffness and debonds the fiber matrix. Nevertheless, nanoclay seems to counteract this effect by minimizing water absorption and enhancing the resistance of the composite to water-soaking degradation [8,29]. This indicates that adding nanoclay can improve the material's durability when exposed to aging factors.

It can be seen from the interaction plot of GF and AG that the tensile strength increases with an increase in glass wt.%. An increase reduces the impact of water-soaking in glass wt.%. With an increase in glass fiber content in fiber-reinforced polymer composites, the influence of water-soaking on tensile strength decreases because of various important factors. Water absorption mainly occurs in the polymer matrix, which is comparatively hydrophilic, while glass fibers are hydrophobic by nature and do not absorb much water. Water absorption also produces swelling and plasticization of the polymer matrix, which reduces its load-carrying capacity. However, with an increased glass fiber fraction, the composite structure remains stiffer and more resistant to such effects, retaining greater mechanical integrity even after extended water exposure [30].

Figure 7 is an interaction plot for the mean flexural strength and a holistic visualization of how nanoclay wt.%, glass fiber wt.%, and age conditions affect the flexural strength of a composite material. The figure comprises several subplots, each showing interactions of

two of these factors at the third factor as a distinguishing parameter. The trends in these plots help interpret how these parameters affect flexural strength together.

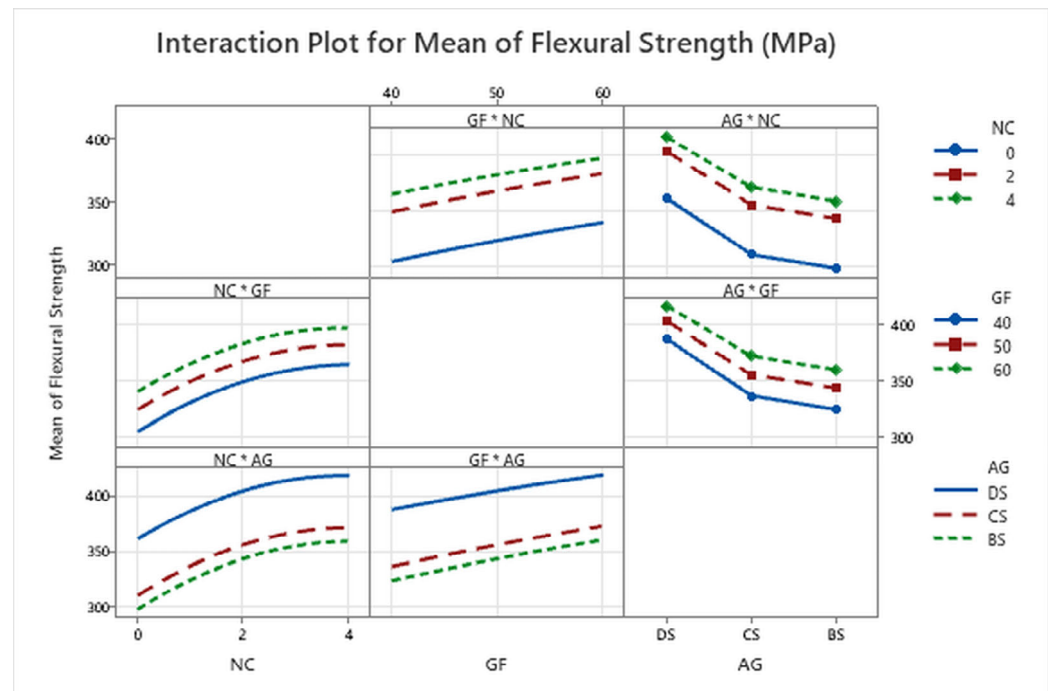


Figure 7. Interaction plot for flexural strength.

The NC and GF interaction plot shows that flexural strength is enhanced with GF's rising content at all NC levels. This is due to the high reinforcing effect of glass fibers, which increase the mechanical properties of the composite. The addition of nanoclay helps to further enhance the composite by enhancing interfacial adhesion at the fiber matrix [31]. However, the enhancement rate in flexural strength is more significant at greater levels of GF, indicating a synergistic interaction between nanoclay and glass fiber reinforcement.

The plot between AG and NC enhances the effect of aging conditions on strength. The strength reduces progressively from DS to CS to BS, as would be expected because of the weakening effects of environmental exposure. The boiling water-soaking exhibits the lowest flexural strength, signifying that exposure to moisture and heat for long periods weakens the composite structure. Nevertheless, incorporating nanoclay retards the weakening process of the composite by serving as a barrier to penetration by moisture and thus minimizing the negative impacts of aging [8].

The AG and GF interaction plot shows that increasing glass fiber content improves flexural strength. Increasing the glass fiber content in fiber-reinforced polymer composites reduces the effect of water-soaking on flexural strength. This is because absorbed moisture lowers the domination of the matrix of the composites on the mechanical properties of composites. However, the reduction in flexural strength across different aging conditions suggests that fiber–matrix adhesion weakens under prolonged exposure to moisture and temperature variations. However, the higher glass fiber contents result in more minor changes in the mechanical properties of composites under aging conditions [30].

3.1.5. Surface Plots

Figures 8 and 9 present surface plots illustrating tensile and flexural strengths in relation to nanoclay and glass fiber wt.%. The aging condition is considered a dry specimen. Also, it can be observed from the figures that the increase in wt.% of glass fiber and nanoclay enhances the strength of the composites.

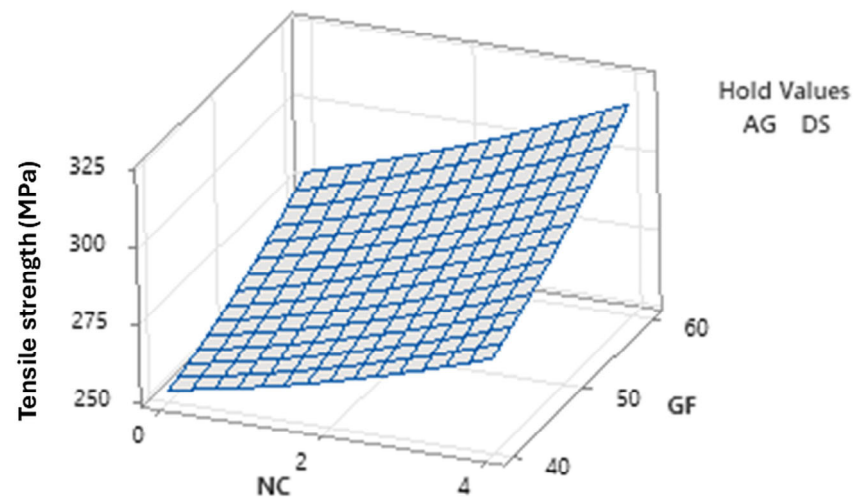


Figure 8. Surface plots for tensile strength.

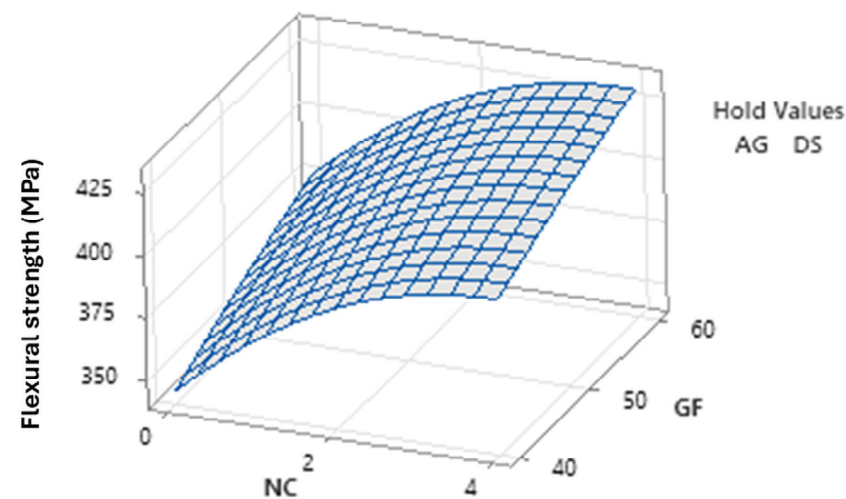


Figure 9. Surface plots for flexural strength.

3.2. Artificial Neural Network

The architecture of the artificial neural network is optimized by varying the number of neurons in a single hidden layer within the range of 5 to 30. The selection of the optimal configuration is based on minimizing the mean squared error (MSE) on the validation set, thereby ensuring a balance between model accuracy and generalization capability. The training is carried out using the Levenberg–Marquardt algorithm, which is well suited for small datasets due to its fast convergence characteristics.

Figure 10 represents the ANN model's mean squared error (MSE) performance curve. The best validation performance is achieved at epoch 23, with an MSE of 0.021256, indicating effective learning and minimal overfitting. The training, validation, and testing curves closely follow each other, suggesting a well-designed model without significant variance issues. The consistent decrease in error during early epochs highlights the model's capacity to adapt the data set effectively, thus stabilizing the error level beyond epoch 20.

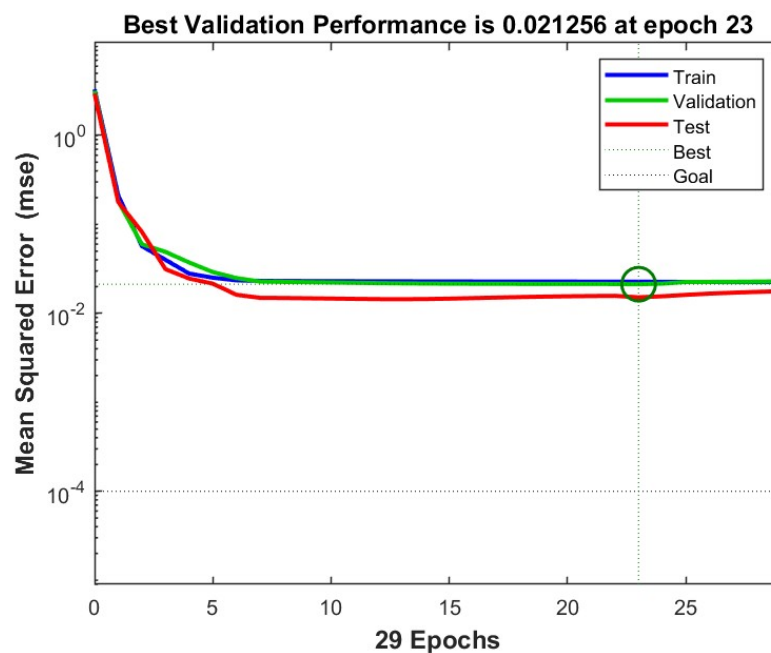


Figure 10. ANN performance plot.

The regression plots in Figure 11 illustrate the correlation between the predicted and the actual values for the training, validation, and test datasets. The correlation coefficient, R -values obtained for the training, validation, and test sets are 0.94365, 0.92366, and 0.96037, respectively, with an overall R -value of 0.94478. These high correlation coefficients indicate a strong agreement between predicted and experimental values, demonstrating the ANN model's ability to learn complex interactions within the dataset.

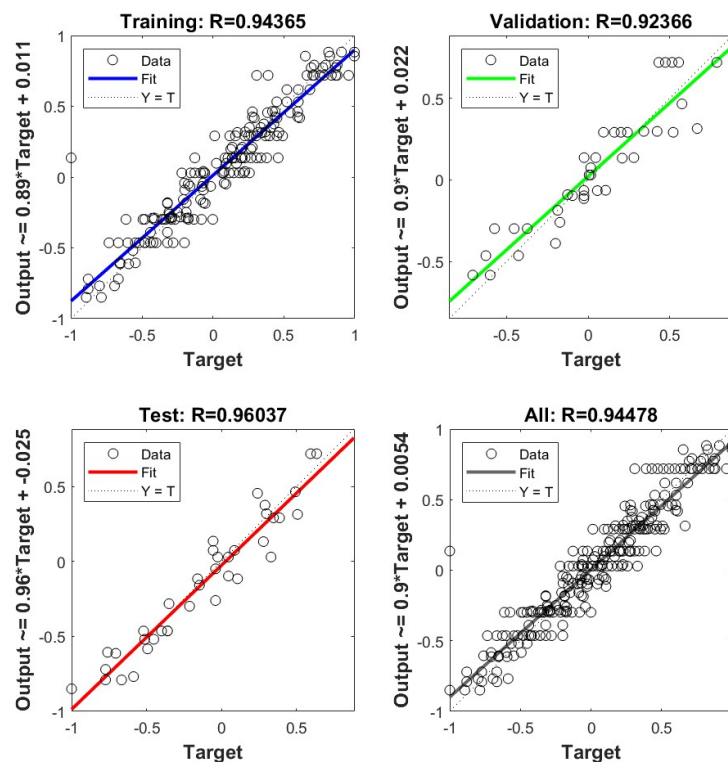


Figure 11. Correlation plots.

The coefficient of determination (R^2) for tensile strength is found to be 0.9620, while for flexural strength, it is 0.9299. Overall, the ANN model exhibits high predictive accuracy

with strong correlation metrics and low validation error, making it a suitable technique for modeling and analyzing the mechanical behavior of glass fiber–nanoclay–epoxy composites.

3.3. Comparison Data

Figure 12 presents the comparison of experimental and RSM- and ANN-predicted values for tensile and flexural strength across multiple trials. The results indicate that the RSM predictions align closely with experimental results for tensile strength, whereas the ANN exhibits better agreement for flexural strength predictions. This further supports the earlier statistical findings where RSM achieved higher accuracy for tensile strength compared to the ANN. The structured polynomial approach of RSM effectively captures the trends in tensile strength due to its relatively linear dependence on the input parameters. On the other hand, the ANN demonstrated superior performance in predicting flexural strength, which can be attributed to its ability to model the intricate dependencies between the input parameters.

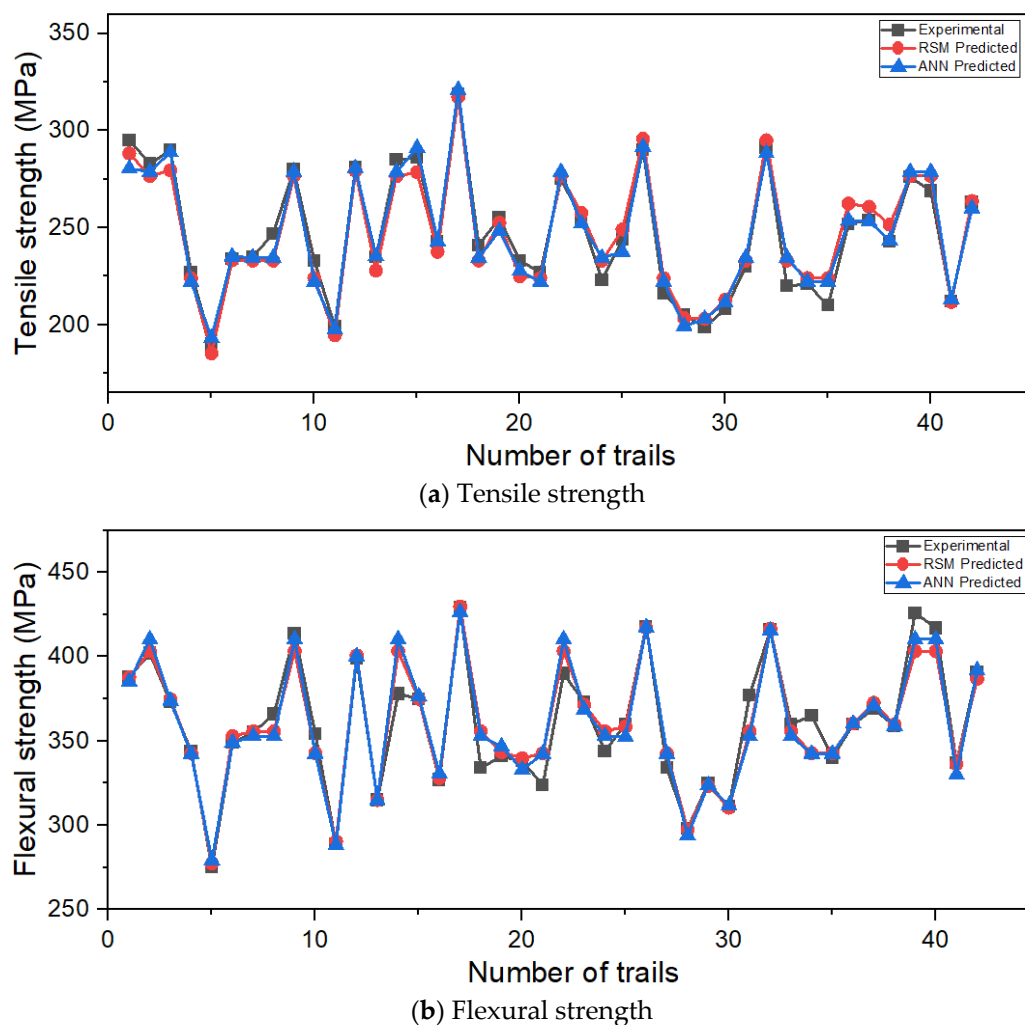


Figure 12. Comparison of experimental findings and the predicted RSM regression values with ANN output.

3.4. SEM Analysis

The SEM images in Figure 13 illustrate the fractured surfaces of composite specimens subjected to tensile loading under different conditions: dry, cold water-soaked, and boiling water-soaked. These images provide insights into the fiber–matrix adhesion and the impact of moisture absorption on the composite’s mechanical performance.

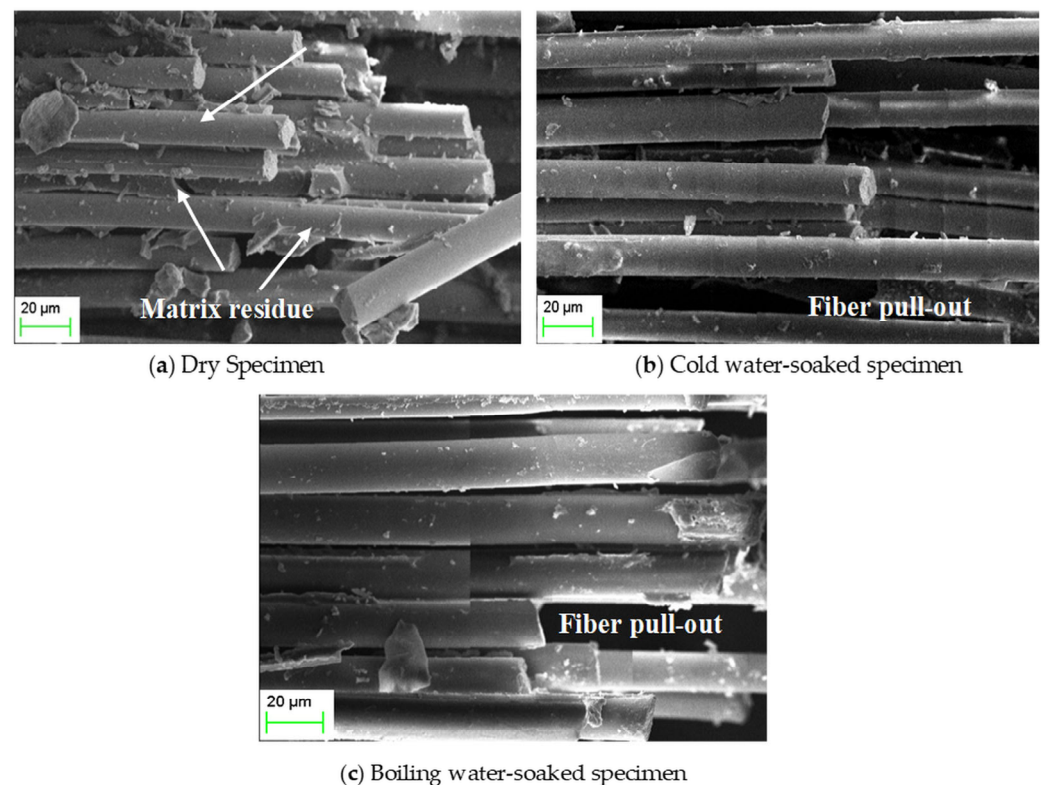


Figure 13. SEM images of the fractured surface under the tensile load of composites without nanoclay.

The fracture surface in the dry specimen (Figure 13a) appears rough and irregular, with noticeable fiber breakage and matrix residue. This indicates a strong fiber–matrix adhesion. The rough surface morphology suggests efficient load transfer between the epoxy matrix and the reinforcing fibers, leading to higher mechanical strength. The presence of fractured fibers rather than fiber pull-out signifies good interfacial bonding, which is crucial for enhancing the composite’s overall durability.

The cold water-soaked specimen (Figure 13b) exhibits a smoother fracture surface with more aligned and exposed fibers. This suggests a reduction in fiber–matrix adhesion due to water absorption. When moisture infiltrates the composite, it weakens the interfacial bonding, leading to fiber pull-out rather than breakage. The boiling water-soaked specimen (Figure 13c) shows the most pronounced fiber exposure, significantly reducing matrix cohesion. Prolonged exposure to high-temperature water likely caused plasticization and hydrolytic degradation of the epoxy matrix, further weakening the fiber–matrix interface. The extensive fiber pull-out observed in this sample suggests severe deterioration, where the matrix is unable to hold the fibers effectively.

The SEM images in Figure 14 display the fractured surfaces of composites with nanoclay under tensile loading. The fracture surface in the dry specimen (Figure 14a) appears rough, with significant fiber breakage and minimal fiber pull-out. This suggests that the addition of nanoclay enhances fiber–matrix bonding, allowing for better load transfer between the epoxy matrix and the reinforcing fibers. Compared to the dry composite specimen without nanoclay (Figure 13a), this specimen exhibits a more compact and cohesive fracture, indicating improved toughness and resistance to brittle failure.

The SEM image (Figure 14b) reveals some exposed fibers for the cold water-soaked specimen but with noticeably less fiber pull-out than the composite without nanoclay (Figure 13b). The boiling water-soaked specimen (Figure 14c) displays more fiber pull-out and matrix deterioration than the dry (Figure 14a) and cold water-soaked (Figure 14b) samples, as high-temperature water accelerates hydrothermal degradation. However,

this specimen retains better structural integrity than the composite without nanoclay (Figure 13c). The reduced fiber pull-out suggests that nanoclay reinforces the matrix, preventing excessive weakening under extreme moisture exposure. This indicates that nanoclay minimizes the influence of water absorption, preserving fiber–matrix adhesion. The matrix degradation is also less pronounced, highlighting the ability of nanoclay to enhance the composite’s resistance to moisture-induced weakening.

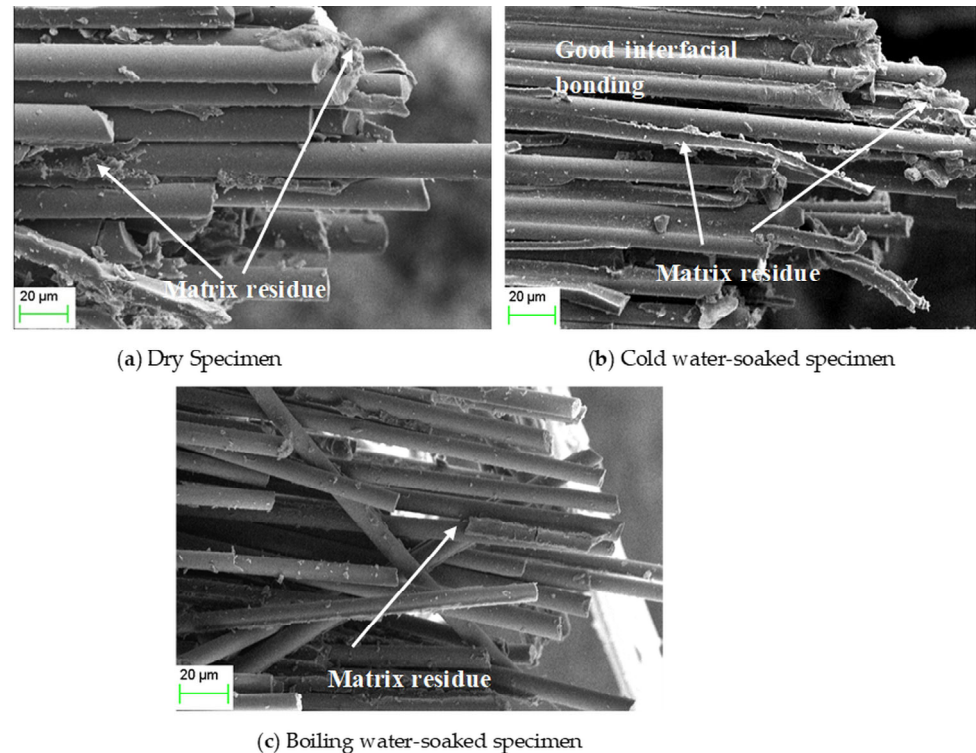


Figure 14. SEM images of the fractured surface under the tensile load of composites with nanoclay.

4. Conclusions

This study investigates the mechanical properties of glass fiber–nanoclay–epoxy composites subjected to different aging conditions using both experimental analysis and predictive modeling approaches. The key findings are summarized as follows:

1. The increase in wt.% of glass fiber and nanoclay enhances the strength of the composites and declines the effect of aging conditions.
2. The composite with 4 wt.% nanoclay and 60 wt.% glass fiber under dry (as-made) conditions displays the highest tensile and flexural strengths, i.e., 319 (± 12) and 429 (± 15) MPa, respectively.
3. The composite with 40 wt.% glass fiber and without nanoclay under boiling water-soaking conditions display the lowest tensile and flexural strengths, i.e., 190 (± 13) and 275 (± 14) MPa, respectively.
4. RSM provided highly accurate predictions for tensile strength due to its structured polynomial approach.
5. The ANN model exhibited enhanced accuracy in predicting flexural strength by effectively modeling the intricate nonlinear relationship.
6. SEM analysis of the fracture surfaces reveals the reasons for specimen failure under tensile load, with apparent differences for dry, cold, and boiling water-soaked specimens.

In summary, this study highlights the effectiveness of combining experimental investigations with predictive modeling to enhance the understanding of composite behavior. The

integration of RSM and ANNs provides a powerful framework for optimizing composite material design.

Author Contributions: Conceptualization, M.S. and A.B.; methodology, M.S. and M.C.G.; software, M.S. and A.B.; validation, M.S., A.B., N.N.K. and M.C.G.; formal analysis, M.S. and A.B.; investigation, M.S. and M.C.G.; resources, N.N.K. and M.C.G.; data curation, M.S. and A.B.; writing—original draft preparation, M.S. and A.B.; writing—review and editing, M.S., A.B., N.N.K. and M.C.G.; visualization, A.B.; supervision, M.C.G.; project administration, N.N.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The data that support the findings of this study are available from the corresponding author Ashwini Bhat, upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Diniță, A.; Ripeanu, R.G.; Ilincă, C.N.; Cursaru, D.; Matei, D.; Naim, R.I.; Tănase, M.; Portoacă, A.I. Advancements in Fiber-Reinforced Polymer Composites: A Comprehensive Analysis. *Polymers* **2023**, *16*, 2. [\[CrossRef\]](#) [\[PubMed\]](#)
2. Hamzat, A.K.; Murad, M.S.; Adediran, I.A.; Asmatulu, E.; Asmatulu, R. Fiber-reinforced composites for aerospace, energy, and marine applications: An insight into failure mechanisms under chemical, thermal, oxidative, and mechanical load conditions. *Adv. Compos. Hybrid Mater.* **2025**, *8*, 152. [\[CrossRef\]](#)
3. Asmare, S.; Yoseph, B.; Jamir, T.M. Investigating the impact resistance of E-glass/Polyester composite materials in variable fiber-to-matrix weight ratio composition. *Cogent Eng.* **2023**, *10*, 2178110. [\[CrossRef\]](#)
4. Mohammadi, H.; Ahmad, Z.; Mazlan, S.A.; Faizal Johari, M.A.; Siebert, G.; Petru, M.; Rahimian Koloor, S.S. Lightweight Glass Fiber-Reinforced Polymer Composite for Automotive Bumper Applications: A Review. *Polymers* **2022**, *15*, 193. [\[CrossRef\]](#)
5. Cheng, W.; Cao, Y. Strength degradation of GFRP cross-ply laminates in hydrothermal conditions. *APL Mater.* **2024**, *12*, 031113. [\[CrossRef\]](#)
6. Thomason, J.; Xypolias, G. Hydrothermal Ageing of Glass Fibre Reinforced Vinyl Ester Composites: A Review. *Polymers* **2023**, *15*, 835. [\[CrossRef\]](#)
7. Cheng, W.; Cao, Y. Investigation of the hygrothermal aging behavior of GFRP laminates used for a marine unmanned aerial vehicle structure. *AIP Adv.* **2023**, *13*, 045107. [\[CrossRef\]](#)
8. Merah, N.; Ashraf, F.; Shaukat, M.M. Mechanical and Moisture Barrier Properties of Epoxy–Nanoclay and Hybrid Epoxy–Nanoclay Glass Fibre Composites: A Review. *Polymers* **2022**, *14*, 1620. [\[CrossRef\]](#)
9. Kiratli, S.; Aslan, Z. Investigation of mechanical properties of nanoclay modified E-glass/epoxy composites. *J. Compos. Mater.* **2023**, *57*, 2501–2512. [\[CrossRef\]](#)
10. Amir, W.W.; Jumahat, A.; Mahmud, J. Effect of nanoclay content on flexural properties of glass fiber reinforced polymer (GFRP) composites. *J. Teknol.* **2015**, *76*, 31–35. [\[CrossRef\]](#)
11. Rafiq, A.; Merah, N. Nanoclay enhancement of flexural properties and water uptake resistance of glass fiber-reinforced epoxy composites at different temperatures. *J. Compos. Mater.* **2019**, *53*, 143–154. [\[CrossRef\]](#)
12. Antil, S.K.; Antil, P.; Singh, S.; Kumar, A.; Pruncu, C.I. Artificial Neural Network and Response Surface Methodology Based Analysis on Solid Particle Erosion Behavior of Polymer Matrix Composites. *Materials* **2020**, *13*, 1381. [\[CrossRef\]](#) [\[PubMed\]](#)
13. Kopparthi, P.K.; Kundavarapu, V.R.; Dasari, V.R.; Kaki, V.R.; Pathakokila, B.R. Modeling of glass fiber reinforced composites for optimal mechanical properties using teaching learning based optimization and artificial neural networks. *SN Appl. Sci.* **2020**, *2*, 131. [\[CrossRef\]](#)
14. Vinoth, V.; Sathiyamurthy, S.; Saravanakumar, S.; Senthilkumar, R. Integrating response surface methodology and machine learning for analyzing the unconventional machining properties of hybrid fiber-reinforced composites. *Polym. Compos.* **2024**, *45*, 6077–6092. [\[CrossRef\]](#)
15. Arunachalam, S.J.; Saravanan, R.; Sathish, T.; Giri, J.; Kanan, M. Mechanical assessment for enhancing hybrid composite performance through silane treatment using RSM and ANN. *Results Eng.* **2024**, *24*, 103309. [\[CrossRef\]](#)
16. Arunachalam, S.J.; Saravanan, R.; Sathish, T.; Alarfaj, A.A.; Giri, J.; Kumar, A. Enhancing mechanical performance of MWCNT filler with jute/kenaf/glass composite: A statistical optimization study using RSM and ANN. *Mater. Technol.* **2024**, *39*, 2381156. [\[CrossRef\]](#)

17. Ladaci, N.; Saadia, A.; Belaadi, A.; Boumaaza, M.; Chai, B.X.; Abdullah, M.M.S.; Al-Khawlani, A.; Ghernaout, D. ANN and RSM Prediction of Water Uptake of Recycled HDPE Biocomposite Reinforced with Treated Palm Waste *W. filifera*. *J. Nat. Fibers* **2024**, *21*, 2356697. [\[CrossRef\]](#)
18. Ahmad, S.M.; Gowrishankar, M.C.; Shettar, M. Effect of boiling water soaking on the mechanical properties and durability of nanoclay-enhanced bamboo and glass fiber epoxy composites. *Sci. Rep.* **2025**, *15*, 3605. [\[CrossRef\]](#)
19. ASTM D3039; Test Method for Tensile Properties of Polymer Matrix Composite Materials. ASTM International: West Conshohocken, PA, USA, 2008. [\[CrossRef\]](#)
20. ASTM D7264; Test Method for Flexural Properties of Polymer Matrix Composite Materials. ASTM International: West Conshohocken, PA, USA, 2021. [\[CrossRef\]](#)
21. Withers, G.J.; Yu, Y.; Khabashesku, V.N.; Cercone, L.; Hadjiev, V.G.; Souza, J.M.; Davis, D.C. Improved mechanical properties of an epoxy glass–fiber composite reinforced with surface organomodified nanoclays. *Compos. Part B Eng.* **2015**, *72*, 175–182. [\[CrossRef\]](#)
22. Jeyakumar, R.; Sampath, P.S.; Ramamoorthi, R.; Ramakrishnan, T. Structural, morphological and mechanical behaviour of glass fibre reinforced epoxy nanoclay composites. *Int. J. Adv. Manuf. Technol.* **2017**, *93*, 527–535. [\[CrossRef\]](#)
23. Annappa, A.R.; Basavarajappa, S.; Davim, J.P. Effect of organoclays on mechanical properties of glass fiber-reinforced epoxy nanocomposite. *Polym. Bull.* **2021**, *79*, 5085–5103. [\[CrossRef\]](#)
24. Çava, K.; İpek, H.; Uşun, A.; Aslan, M. Examine the mechanical properties of woven glass fiber fabric reinforced composite plate manufactured with vat-photopolymerization. *Polym. Compos.* **2024**, *45*, 17105–17120. [\[CrossRef\]](#)
25. Shettar, M.; Kini, U.A.; Sharma, S.; Hiremath, P.; Gowrishankar, M.C. Hygrothermal chamber aging effect on mechanical behavior and morphology of glass fiber-epoxy-nanoclay composites. *Mater. Res. Express* **2020**, *7*, 015318. [\[CrossRef\]](#)
26. Borges, C.S.P.; Akhavan-Safar, A.; Marques, E.A.S.; Carbas, R.J.C.; Ueffing, C.; Weißgraeber, P.; da Silva, L.F.M. Effect of Water Ingress on the Mechanical and Chemical Properties of Polybutylene Terephthalate Reinforced with Glass Fibers. *Materials* **2021**, *14*, 1261. [\[CrossRef\]](#)
27. Kini, U.A.; Shettar, M.; Sharma, S.; Hiremath, P.; Gowrishankar, M.C.; Hegde, A.; Siddhartha, D. Effect of hygrothermal aging on the mechanical properties of nanoclay-glass fiber-epoxy composite and optimization using full factorial design. *Mater. Res. Express* **2019**, *6*, 065311. [\[CrossRef\]](#)
28. Bin Rashid, A.; Haque, M.; Islam, S.M.M.; Labib, K.R.U. Nanotechnology-enhanced fiber-reinforced polymer composites: Recent advancements on processing techniques and applications. *Heliyon* **2024**, *10*, e24692. [\[CrossRef\]](#)
29. Merah, N.; Mohamed, O. Nanoclay and Water Uptake Effects on Mechanical Properties of Unsaturated Polyester. *J. Nanomater.* **2019**, *2019*, 8130419. [\[CrossRef\]](#)
30. Chaichanawong, J.; Thongchuea, C.; Areerat, S. Effect of moisture on the mechanical properties of glass fiber reinforced polyamide composites. *Adv. Powder Technol.* **2016**, *27*, 898–902. [\[CrossRef\]](#)
31. Nemati Giv, A.; Rastegar, S.; Özcan, M. Influence of nanoclays on water uptake and flexural strength of glass–polyester composites. *J. Appl. Biomater. Funct. Mater.* **2020**, *18*, 228080002093018. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.